1 Probability Review

1.1 Probability Concept and Definitions

What is probability conceptually?

The term *probability* may be conceptualized as a measure of one's belief in the occurrence of a future event. The concept of probability is indispensable in work with physical, biological or social mechanisms that generate observations that cannot be predicted with certainty. For example,

- it is not possible to know the exact load that a bridge will endure before collapsing into a river
- the blood pressure of a person at a given point of time cannot be predicted with certainty
- it is not possible to predict the occurrence of heads on a single coin of a balanced coin

Random events

All three events given in the above example are called *random* or *stochastic* events. In classical definition of probability, a random event is a part of all possible outcomes obtained from a random experiment. Note that the set of all possible outcomes is called a sample space. Set theoretic notions are applied to define the probability of a random event.

History of Probability

Historically the theory of probability has been associated with gambling, and most textbook examples still come from that activity. Although gambling dates back thousands of years, the birth of modern probability is considered to be a 1654 letter from the Flemish aristocrat and notorious gambler Chevalier de Mere to the mathematician and philosopher Blaise Pascal. Later correspondence between Pascal and Pierre de Fermet led to the formal development of probability theory. ## Three definitions of Probability In the literature,

there are three types of probability definitions as stated below:

- Classical or theoretical definition of probability
- Empirical or relative frequency definition of probability
- Subjective probability

1.1.1 Classical or theoretical definition of probability

The theoretical definition is the most precise approach based on mathematical theories including set theory and measure theory. In this approach all possible outcomes of a random experiment are assumed to be known. To calculate probability of an outcome or an event (set of outcomes), the possible outcomes are either

assumed to be equally likely or a mathematical measure is assigned to each of these outcomes.

If all possible outcomes are assumed to be equally likely, the classical definition of the probability of an event E is given as,

$$P(E) = \frac{n(E)}{n(S)},$$

where $n(E)$ is the number of outcomes in the event, and $n(S)$ is the number of outcomes in the sample space S. Note that $P(E)$ denotes that *the probability that E occurs.* ### 1.1.2 Empirical or relative frequency definition of probability The empirical probability of an event is calculated by dividing the number of times the event occurs by the total number of trials performed. Thus the empirical probability of an event E is defined as

$$P(E) = \frac{f}{n},$$

where f is the frequency of event E and n is the number of times the experiment is performed. This definition is also known as the relative frequency definition of the probability of an event E. *Note:* in this course we will primarily adopt this definition to define probability of an event due to its intuitive approach. ### Subjective probability Subjective probability is simply an educated guess about the chance that an event will occur. It is usually derived from an individual's personal judgment or own experience about whether a specific outcome is likely to occur. It contains no formal calculations and only reflects the subject's opinions and past experience.

1.1.4 Examples

Example 1. Theoretical probability: Toss a fair coin three times. What is the probability of seeing one head in these tosses? Here the random experiment is tossing a fair coin, since outcome of a specific trial is uncertain but all possible outcomes are known. So the sample space can be written as follows:

$$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$$

Note that the sample space S is a set that contains 8 eqully likely outcomes. The event of seeing a head can be written as $E = \{HTT, THT, TTH\}$. Then

$$P(E) = \frac{n(E)}{n(S)} = \frac{3}{8}.$$

Example 2. Empirical probability: In this example you will see the implication of relative frequency to obtain the probability of head when a fair coin is tossed a large number of time. First we will simulate the coin tossing using R, using the function sample().

```r
sample.space <- c(0,1) # 0 <- tail, 1 <- head
theta <- 0.5 # this is a fair coin
N <- 100000 # we want to flip a coin 100000 times

flips <- sample(sample.space,
                size = N,
                replace = TRUE,
                prob = c(theta, 1 - theta))
prop.table(table(flips))
```

```
## flips
##       0       1
## 0.49848 0.50152
```

Note that the relative frequency of head when you toss a fair coin large number of times (10^5) is close to 0.5. As you increase the size, the relative frequency will become very close to the theoretical probability which is 0.5.

Example 3. Subjective probability: An example of subjective probability is a "gut instinct'' when making a trade for a stock. If you believe that the market index will go up, you are more likely to make a trade in the direction of selling your stock. Here you are using your educated guess that the chance of making profit from selling the stock is higher. ## 1.1.5 Real Life Examples of applications of probability **Example 1:**

Weather Forecasts. Before planning for an outing, we check the weather forecast. Suppose it says that there is a 70% chance that it may rain. Where does this 70% come from? Meteorologists use probability models to predict the weather forecast. They look at the historical database of the days, which have similar characteristics of temperature, humidity, and pressure, etc. And determine that on 70 out of 100 similar days in the past, it had rained. **Example 2: Lottery Tickes**. Winning or losing a lottery is an interesting

example of probability. In a typical Lottery game, each player chooses six distinct numbers from a particular range. If all the six numbers on a ticket match with that of the winning lottery ticket, the ticket holder is a Jackpot winner- regardless of the order of the numbers. The probability of this happening is 1 out of a million.

1.2 Counting Principles and Combinatorics

Combinatorics based computation for finding classical probability

Recall, in classical definition of probability of an event all the outcomes are assumed to be equally likely. The computation of the probability can be handled easily with counting all the outcomes of interest and dividing by the total number of possible outcomes. Thus counting principles, or combinatorics play an important role in probability computation. Four commonly used results from combinatorics are presented as follows:

1. The mn rule: With m elements $a_1, a_2, \ldots, a_m$ and n elements $b_1, b_2, \ldots, b_n$ it is possible to form $mn = m \times n$ pairs containing one element from each group. Note that the mn rule can be extended to any number of sets. Given three sets of elements $a_1, a_2, \ldots, a_m$, $b_1, b_2, \ldots, b_n$, and $c_1, c_2, \ldots, c_p$ the number of distinct triplets containing one element from each set is $nmp = m \times n \times p$.

2. Permutation: An ordered arrangement of r distinct objects is called a *permutation*. The number of ways

of ordering n distinct objects taken r at a time is denoted by P_r^n and is defined as

$$P_r^n = n(n-1)(n-2)\dots(n-r+1) = \frac{n!}{(n-r)!}$$

3. Combination: The number of combinations of n objects taken r at a time is the number of subsets, each of size r that can be formed from the n objects. This number is denoted by C_r^n. Thus the combination is the number unordered subsets of size r chosen without replacement from n available objects and is given by,

$$C_r^n = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

Note that the combination can be viewed as partitioning the n objects into two groups, the r selected and the $(n-r)$ remaining. This idea can be extended to partitioning n distinct objects into k distinct groups containing $n_1, n_2, \dots, n_k$ objects respectively and leads to the fourth rule.

4. Multinomial coefficient: The number of ways of partitioning n distinct objects into k distinct groups containing $n_1, n_2, \dots, n_k$ objects, respectively, where each object appears in exactly one group and $\sum_{i=1}^{k} n_i = n$ is

$$\binom{n}{n_1 n_2 \dots n_k} = \frac{n!}{n_1! n_2! \dots n_k!}$$

The terms $\binom{n}{n_1 n_2 \dots n_k}$ are often called multinomial coefficients because they occur in the expansion of the multinomial term $y_1 + y_2 + \dots + y_k$ raised to the nth power:

$$(y_1 + y_2 + \dots + y_k)^n = \sum \binom{n}{n_1 n_2 \dots n_k} y_1^{n_1} y_2^{n_2} \dots y^{n_k}$$

where the sum is taken over all $n_i = 0, 1, \dots, n$ such that $n_1 + n_2 + \dots + n_k = n$.

1.2.1 Examples

Example 1: The names of three employees are to be drawn, without replacement, from a bowl containing the names of 30 employees of a small company. The person whose name is drawn first receives \$100, and the individuals whose names are drawn second and third receive \$50 and \$25, respectively. How many sample points are associated with this experiment? **Solution**: Because the prizes awarded are different, the number of sample points is the number of ordered arrangements of $r = 3$ out of the possible $n = 30$ names. Thus the total number of samples points is

$$P_3^{30} = \frac{30!}{27!} = 24,360.$$

Example 2: Which is more likely in five cards: one king or two hearts? **Solution**:

$$P(1 \text{ king}) = \frac{\binom{4}{1} \times \binom{48}{4}}{\binom{52}{5}} = 0.299$$

$$P(\text{2 hearts}) = \frac{\binom{13}{2} \times \binom{39}{3}}{\binom{52}{5}} = 0.274$$

```
choose(4,1)*choose(48,4)/choose(52,5)
```

```
## [1] 0.2994736
```

```
choose(13,2)*choose(39,3)/choose(52,5)
```

```
## [1] 0.2742797
```

Example 3: Probability of getting four aces in a bridge hand A standard deck of 52 cards is dealt to four players, 13 cards each. One of the players is Millie. What is the probability that Millie is dealt all four aces? Note there are

$$\frac{52!}{13!13!13!13!}$$

possible deals. The number of deals in which Millie holds all four aces is

$$\frac{48!}{13!13!13!9!}.$$

Thus the required probability is:

$$\frac{\frac{48!}{13!13!13!9!}}{\frac{52!}{13!13!13!13!}} = 0.00264.$$

1.3 Probability Laws

1.3.1 Properties of probability

- For any event E, $0 \leq P(E) \leq 1$
- For any sample space, $S, P(S) = 1$
- For any null event, which is an empty set $\Phi, P(\Phi) = 0$
- The sum of the probabilities of an event E and its complement E^c is 1. That is $P(E) + P(E^c) = 1$, implying $P(E^c) = 1 - P(E)$

1.3.2 Multiplicative Law of Probability

The intersection of two events E_1 and E_2 is denoted as $E_1 \cap E_2$ implying occurrence of two events E_1 and E_2 simultaneously. The probability of the intersection of two events E_1 and E_2 is,

$$P(E_1 \cap E_2) = P(E_1)P(E_2|E_1) = P(E_2)P(E_1|E_2)$$

If E_1 and E_2 are independent then

$$P(E_1 \cap E_2) = P(E_1)P(E_2)$$

The multiplication law follows directly from the definition of conditional probability

The multiplication law can be extended for more than two events. For three events, E_1, E_2 and E_3

$$P(E_1 \cap E_2 \cap E_3) = P(E_1|E_2, E_3)P(E_2|E_3)P(E_3)$$

If If E_1, E_2 and E_3 are independent then

$$P(E_1 \cap E_2 \cap E_3) = P(E_1)P(E_2)P(E_3)$$

1.3.3 Additive Law of Probability

The union of two events E_1 and E_2 is denoted as $E_1 \cup E_2$ implying occurrence of either event E_1 or event E_2 or both. The probability of the union of two events E_1 and E_2 is,

$$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$$

If E_1 and E_2 are disjoint or mutually exclusive then $P(E_1 \cap E_2) = 0$ and

$$P(E_1 \cup E_2) = P(E_1) + P(E_2)$$

Note: Two events are disjoint or mutually exclusive if it is impossible for them to occur together.

The additive law can be extended for more than two events. For three events, E_1, E_2 and E_3,

$$P(E_1 \cup E_2 \cup E_3) = P(E_1) + P(E_2) + P(E_3) - P(E_1 \cap E_2) - P(E_1 \cap E_3) - P(E_2 \cap E_3) + P(E_1 \cap E_2 \cap E_3)$$

If E_1, E_2 and E_3 are disjoint or mutually exclusive then

$$P(E_1 \cup E_2 \cup E_3) = P(E_1) + P(E_2) + P(E_3)$$

1.3.4 Examples

Please check section 1.5 for a detail example on `The Bus ridership model` for applications of additive and multiplicative laws of probabilities.

1.4 Conditional Probability

The probability of an event sometimes depends upon whether we know that other events have occurred. For example, we may be interested in the probability that it will rain tomorrow given that it had been raining for two days. Here the event that it will rain tomorrow depends on the weather condition of previous two days. The probability of such events are called *conditional probability.*

The *conditional probability* of an event E_1 given that an event E_2 has occurred is defined as

$$P(E_1|E_2) = \frac{P(E_1 \cap E_2)}{P(E_2)},$$

provided $P(E_2) > 0$. The notation $P(E_1|E_2)$ is read as " probability of E_1 given E_2' '. ## 1.4.1 Stochastic

Independence When the probability of an event E_1 is unaffected by the occurrence or nonoccurrence of

event E_2, we say that events E_1 and E_2 are independent. Events E_1 and E_2 are said to be stochastically independent if one of the followings holds:

$$P(E_1|E_2) = P(E_1)$$
$$P(E_2|E_1) = P(E_2)$$
$$P(E_1 \cap E_2) = P(E_1)P(E_2).$$

Otherwise the events are said to be dependent.

1.5 Examples

1.5.1 Probability Laws and Conditional Probability Examples - Bus Ridership Case Study

This case study has been adapted from the `Bus Ridership` example in one of the recommended text `Probability and Statistics for Data Science` by Norman Matloff.

In a simplified version of a bus ridership model that could be used by the bus company/agency to plan the number of buses and the frequency or stops we assume the following:

- At each stop, each passenger alights from the bus, independently of the actions of others, with probability 0.2 each.
- Either 0,1 or 2 new passengers get on the bus, with probabilities 0.5, 0.4, and 0.1, respectively. Passengers at successive stops act independently.
- Assume the bus is so large that it never becomes full, so new passengers can always board.
- Assume that the bus is empty when it arrives at the first stop.

Notations: Let L_i denote the number of passengers on the bus as it leaves its ith stop, $i = 1, 2, 3...$ Let B_i denote the number of new passengers who board the bus at the ith stop.

1. What is the probability that no passenger board at the first three stops?

   Note here, $B_1 = 0$, $B_2 = 0$, $B_3 = 0$ Applying multiplication rule for independent events you can find

the desired probability:

$$\begin{aligned} P(B_1 = 0 \text{ and } B_2 = 0 \text{ and } B_3 = 0) &= P(B_1 = 0) \times P(B_2 = 0) \times P(B_3 = 0) \\ &= 0.5 \times 0.5 \times 0.5 = 0.125 \end{aligned}$$

2. What is the probability that the bus leaves the second stop empty? Note here we want $P(L_2 = 0)$

$$\begin{aligned} L_2 = 0 \quad \approx \quad & B_1 = 0, \quad L_2 = 0 \\ & B_1 = 1, \quad L_2 = 0 \\ & B_1 = 2, \quad L_2 = 0 \end{aligned}$$

Then,

$$\begin{aligned} P(L_2 = 0) &= P\left(\{B_1 = 0 \quad \& \quad L_2 = 0\} \text{ or } \{B_1 = 1 \quad \& \quad L_2 = 0\} \text{ or } \{B_1 = 2 \quad \& \quad L_2 = 0\}\right) \\ &= \sum_{i=0}^{2} P(B_1 = i \quad \& \quad L_2 = 0) \\ &= \sum_{i=0}^{2} P(B_1 = i) P(L_2 = 0 | B_1 = i) \\ &= P(B_1 = 0) P(L_2 = 0 | B_1 = 0) + P(B_1 = 1) P(L_2 = 0 | B_1 = 1) + P(B_1 = 2) P(L_2 = 0 | B_1 = 2) \\ &= 0.5 \times 0.5 + 0.4 \times 0.2 \times 0.5 + 0.1 \times (0.2)^2 \times 0.5 = 0.292 \end{aligned}$$

Note: $L_2 = 0 | B_1 = 1 \approx 1$ leaves at stop 2 wp 0.2 & none board at stop 2 wp 0.5 $L_2 = 0 | B_2 = 2 \approx 2$

leaves at stop 2 wp $(0.2)^2$ & none board stop 2 wp 0.5.

3. Given the bus arrives empty at the third stop, what is the probability that exactly 2 people boarded at the first stop? Note this is a conditional probability that we would like to find.

$$\begin{aligned} P(B_1 = 2 | L_2 = 0) &= \frac{P(B_1 = 2 \quad \& \quad L_2 = 0)}{P(L_2 = 0)} \\ &= \frac{P(B_1 = 2) P(L_2 = 0 | B_1 = 2)}{P(L_2 = 0)} \\ &= \frac{0.1 \times (0.2)^2 \times (0.5)}{0.292} \simeq 0.0068 \end{aligned}$$

4. What is the probability that fewer people board at second stop? Here we want to find $P(B_2 < B_1)$

Note:

$$\begin{aligned} B_2 < B_1 \quad \approx \quad & B_1 = 1, \quad B_2 < B_1 \text{ or} \\ & B_1 = 2, \quad B_2 < B_1 \end{aligned}$$

$$\begin{aligned}\therefore \quad P(B_2 < B_1) &= P\left(\{B_1 = 1 \quad \& \quad B_2 < B_1\} \text{ or } \{B_1 = 2, \quad B_2 < B_1\}\right) \\ &= P(B_1 = 1, \quad B_2 < B_1) + P(B_1 = 2, \quad B_2 < B_1) \\ &= 0.4 \times 0.5 + 0.1 \times (0.5 + 0.4) = 0.29\end{aligned}$$

Note $B_1 = 1, \quad B_2 < B_1 \implies$ 1 boards at first stop and with probability 0.4 and 0 boards at the second stop with probability 0.5. Similarly, $B_1 = 2, \quad B_2 < B_1 \implies$ 2 board at first stop and with probability 0.1 and either 0 or 1 boards at the second stop with probability (0.5+0.4).

5. Given that a passenger got off the bus at the second stop and that the bus leaving that stop empty, what is the probability that the person was the only passenger when the bus left the first stop? Note

   we want to find $P(L_1 = 1 | L_2 = 0, L_1 > 0)$

$$\begin{aligned}P(L_1 = 1 | L_2 = 0, L_1 > 0) &= \frac{P(L_1 = 1, L_2 = 0, L_1 > 0)}{P(L_2 = 0, L_1 > 0)} \\ &= \frac{P(L_1 = 1, L_2 = 0)}{P(L_2 = 0, L_1 > 0)}\end{aligned}$$

   Note:

$$\begin{aligned}P(L_2 = 0, L_1 > 0) &= P(L_2 = 0, L_1 = 1) + P(L_2 = 0, L_1 = 2) \\ &= P(B_1 = 1, L_2 = 0) + P(B_1 = 2, L_2 = 0) = 0.4 \times 0.2 \times 0.5 + 0.1 \times (0.2)^2 \times 0.5 = 0.042\end{aligned}$$

   $\therefore$ The required probability is:

$$\frac{0.4 \times 0.2 \times 0.5}{0.042} \simeq 0.95$$

1.6 Bayes' Rule

1.6.1 What is Bayes' theorem?

Bayes' theorem, named after 18th-century British mathematician Thomas Bayes, is a mathematical formula for determining conditional probabilities. Bayes' theorem provides a way to update probabilities given new or additional evidence. Note that the conditional probability $P(A|B)$ can be expressed as follows:

$$P(A|B) = \frac{P(A)P(B|A)}{P(B)} = \frac{P(A)P(B|A)}{P(A)P(B|A) + P(A^c)P(B|A^c)}$$

Note, $P(B) = P(A)P(B|A) + P(A^c)P(B|A^c)$ is called the total law of probability and can be extended for more than two partitions of the sample space. Here, $P(A)$ and $P(A^c)$ are called the prior probabilities, and the updated probability $P(A|B)$ is called the posterior probability.

1.6.2 Bayes' Rule Example: Defective fuses in a production line

An electronic fuse is produced by five production lines in a manufacturing operation. These fuses are costly, reliable, and are shipped to suppliers in 100-unit lots. Because testing is destructive, most buyers of the fuses test only a small number of fuses before deciding to accept or reject lots of incoming fuses.

All five production lines produce fuses at the same rate and normally produce only 2% defective fuses, which are dispersed randomly in the output. However, production line 1 suffered mechanical difficulty and produced 5% defectives during the month of March. This situation became known to the manufacturer after the fuses had been shipped.

A customer received a lot produced in March and tested three fuses; one failed.

- What is the probability that the lot was produced on line 1?

- What is the probability that the lot came from one of the four other lines?

Let L_i denote a fuse that was drawn from line $i, i = 1, 2, 3, 4, 5$ and let D denote the event that a fuse was defective.

$$P(L_1) = 0.2,$$

which is the probability that line produces the fuse. Note $P(L_1)$ is an unconditional probability.

$$P(D|L_1) = 0.05(0.95)^2 + 0.05(0.95)^2 + 0.05(0.95)^2 = 3(0.05)(0.95)^2 = 0.135375,$$

which is a conditional probability and is computed first applying the addition rule (any of the three fuses can be defective) and multiplication rule on 1 defective and 2 non-defectives.

Similarly,

$$P(\bar{L_1}) = 0.8$$

, where $\bar{L_1}$ represent that the fuse was not drawn from line 1.

$$P(D|\bar{L_1}) = 3(0.02)(0.98)^2 = 0.057624.$$

Then

$$\begin{aligned} P(D) = P(\{D, L_1\} \text{ or } \{D, \bar{L_1}\}) = P(L_1)P(D|L_1) + P(\bar{L_1})P(D|\bar{L_1}) \\ = (0.2) * (0.135375) + (0.8) * (0.057624) = 0.0731742 \end{aligned}$$

Finally,

$$P(L_1|D) = \frac{P(L_1, D)}{P(D)} = \frac{P(L_1)P(D|L_1)}{P(D)} = \frac{(0.2) * (0.135375)}{0.0731742} = 0.37,$$

and

$$P(\bar{L_1}|D) = 1 - P(L_1|D) = 1 - 0.37 = 0.63.$$

1.6.3 Bayes' Rule Example: Document classification

In the field of *text classification*, one of the objectives is to determine the topic categories of new documents. Say for example a software for this task sees a document that contains the word *bonds*. If we also know that the document contains the word *interest*, the Bayes' rule can be applied to find the probability,

$$P(\text{financial}|\text{bonds,interest}).$$

This probability can be estimated from a dataset of documents, known as a *corpus*, as the proportion of financial documents among all documents containing the two specified words.

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